

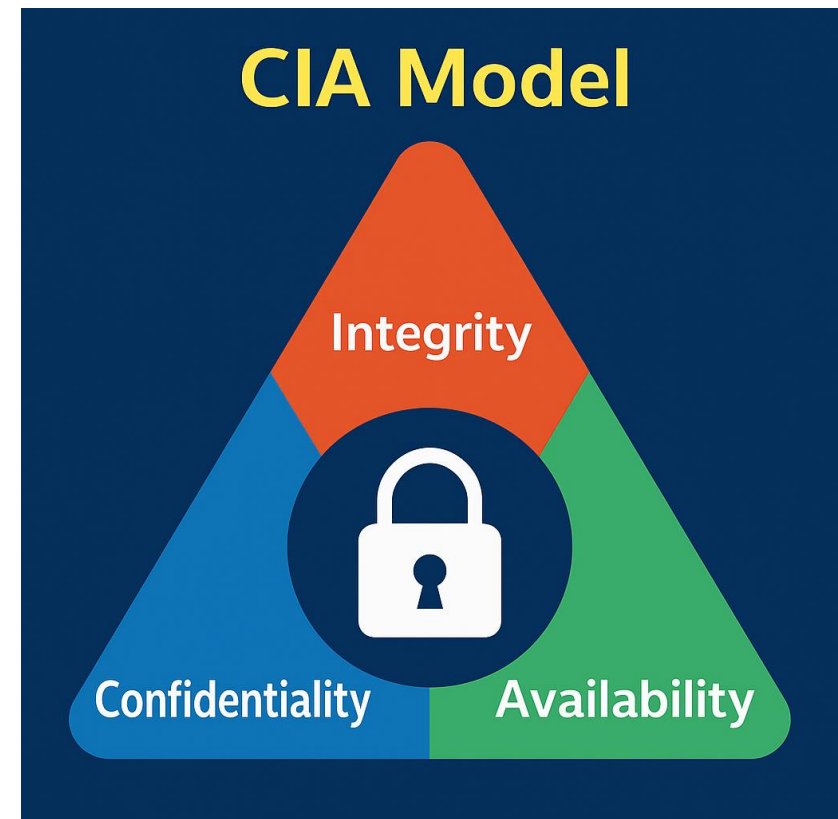
Network security

- refers to the practices, policies, and technologies that protect a computer network and its data from:
 - unauthorized access,
 - misuse,
 - modification,
 - or denial of service.
- using hardware and software tools, such as firewalls and antivirus, along with processes and procedures to safeguard a network's integrity, confidentiality, and availability from cyberattacks and threats.

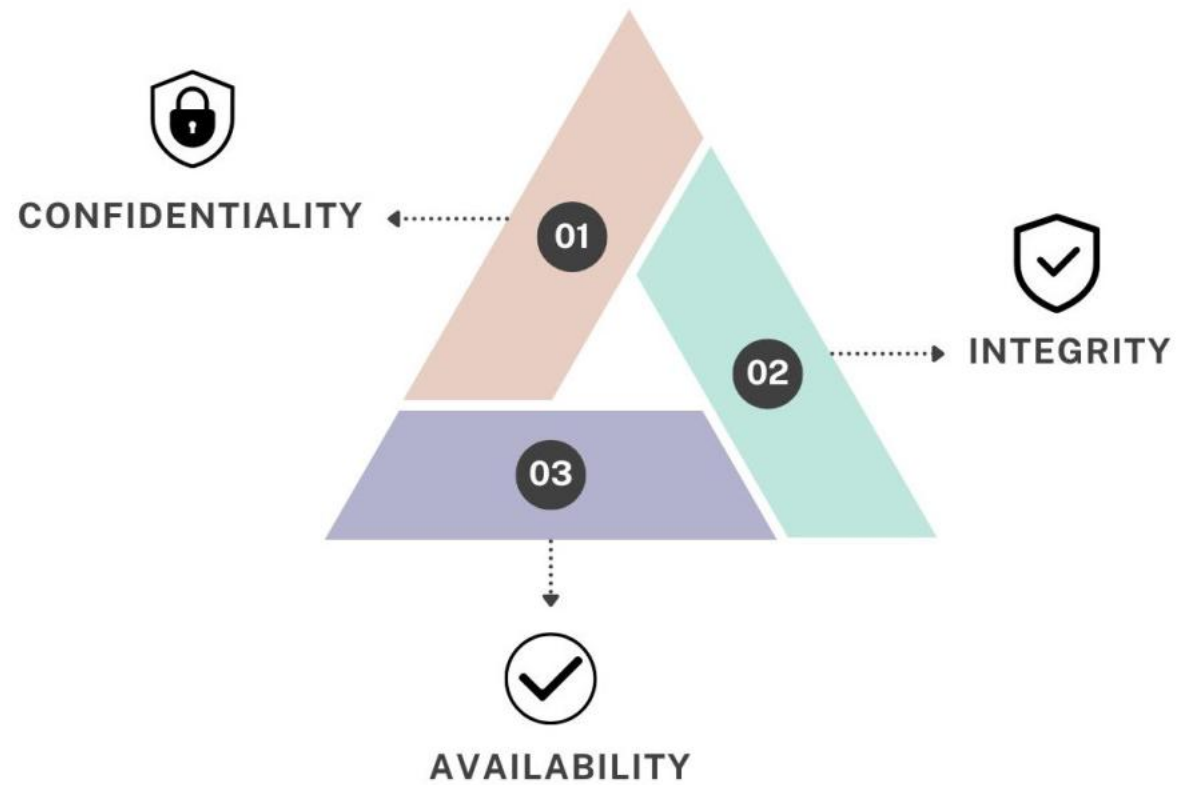


What is Network Security?

- Refers to protecting data, devices, and networks from unauthorized access or attacks.
- Ensures Confidentiality, Integrity, and Availability (CIA) of information.



CIA Triad



CIA Triad

- Network security centers on confidentiality, integrity, and availability of the company's network infrastructure—the so-called “CIA triad.”
- **Confidentiality** means that sensitive information stays in the right hands, blocking unauthorized access.
- **Integrity** is what keeps data accurate and consistent, preventing tampering or unintended changes during storage or transmission.
- **Availability** assures network services and data are accessible when authorized users need them.

CIA Summary

Term	Focus	Main Threat	How to Protect
Confidentiality	Keep secrets safe	Unauthorized access	Encryption, authentication
Integrity	Keep data correct	Data alteration	Hashing, permissions
Availability	Keep data usable	Downtime, attacks	Backups, redundancy

Importance of Network Security

- Protects sensitive data.
- Prevents identity theft.
- Ensures business continuity.
- Builds trust in digital systems.

Common Security Threats

Threat Type	Description	Example
Virus	Infects files and replicates	Infected attachments
Worm	Self-replicating over networks	Email-spreading worm
Trojan Horse	Disguised as safe software	Fake antivirus
Spyware	Monitors user activity	Keylogger
Phishing	Tricking users to share info	Fake bank email
Ransomware	Locks data until payment	WannaCry attack

Cyber Attack Types

- Denial of Service (DoS): floods a server with requests.
- Man-in-the-Middle (MitM): intercepts data between sender and receiver.
- Password Attacks: brute force or guessing passwords.
- Social Engineering: manipulation to gain access.



Network Security Tools

Tool	Function
Firewall	Blocks unauthorized access.
Antivirus	Detects and removes malware.
Encryption	Secures data during transmission.
VPN (Virtual Private Network)	Encrypts traffic and hides identity.
IDS/IPS	Detects or prevents intrusion.

Encryption Concept

- Converts data into unreadable form (cipher text).
- Only authorized users can decrypt it.
- Example:
 - Plain text → Encryption → Cipher text → Decryption → Plain text.

Stage	Message	Readable?
Plain Text	HELLO	 Yes
After Encryption (Cipher Text)	JGNNQ	 No
After Decryption	HELLO	 Yes

Safe Internet Practices

- Brainstorm ??

Safe Internet Practices

- Use strong, unique passwords.
- Keep software updated.
- Avoid clicking unknown links.
- Use secure (HTTPS) websites.
- Log out from public computers.

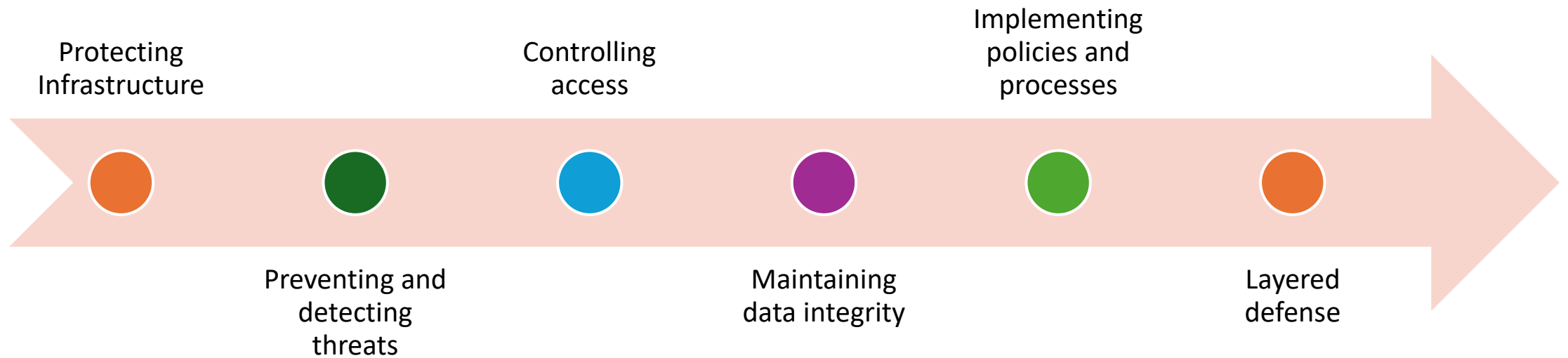
Safe Internet Practices

- **Use strong, unique passwords and multi-factor authentication (MFA):** Create long passwords with a mix of characters (letters, numbers, symbols) and use a password manager for organization. Enable MFA for an extra layer of security on your accounts.
- **Protect personal information:** Avoid sharing sensitive details like your home address, phone number, or social security number with strangers or on public platforms.
- **Be cautious of suspicious links and downloads:** A careless click can lead to malware or data exposure. Avoid clicking links from untrusted sources, opening attachments in spam emails, or downloading content from unofficial websites.
- **Secure your devices and network:** Use passcodes, fingerprint readers, or face-scanning technology on your devices. Use a secure internet connection (HTTPS) and a VPN when using public Wi-Fi.
- **Keep software updated:** Regularly install the latest security updates for your operating systems and apps, as they often contain patches for vulnerabilities.
- **Back up your data:** Regularly back up important files to an external hard drive or cloud storage to protect against data loss from hardware failure or ransomware attacks.

Communication Ethics

- Respect privacy and intellectual property.
- Avoid spreading misinformation.
- Be polite and responsible in online discussions.
- Report cyberbullying or unethical content.
- Think before you post
- Be honest and authentic
- Respect intellectual property and copyright laws
- Do not pretend to be someone else or create fake profiles to fool others
- Respect Privacy
- Ethical online communication (often called "netiquette") requires individuals to be mindful of their interactions and the impact of their words and actions on others.

Key aspects of network security



Key aspects of network security [Additional]

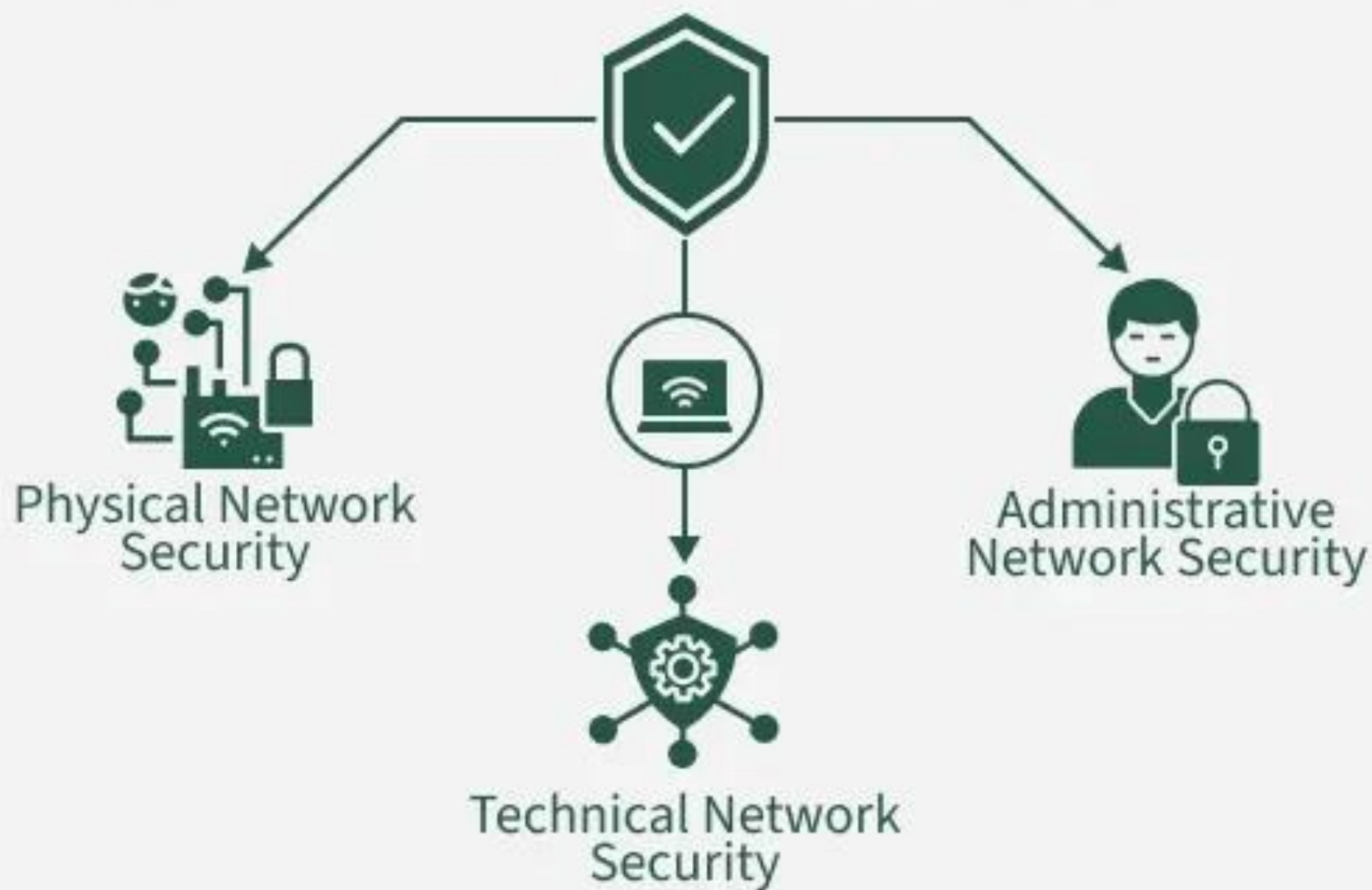
- **Protecting infrastructure:** It secures the underlying networking infrastructure, including hardware, software, and data centers (both physical and cloud-based).
- **Preventing and detecting threats:** It uses tools like firewalls, intrusion prevention systems (IPS), and antivirus software to prevent and detect malicious activities and attacks.
- **Controlling access:** It involves implementing access control measures to manage who can access the network and its resources.



Key aspects of network security [Additional]

- **Maintaining data integrity:** Its goal is to ensure the data transmitted across the network remains safe and secure, protecting it from being compromised.
- **Implementing policies and processes:** It includes establishing policies and processes for network use, management, and overall threat protection.
- **Layered defense:** Effective network security uses a layered defense that combines physical, technical, and administrative components

HOW DOES NETWORK SECURITY WORK?



How Does Network Security Work?

- Network security uses multiple layers of protection, both at the network edge and within the internal environment



How Does Network Security Work?

1. **Physical Network Security:** It is the foundational level of network protection, focused on preventing unauthorized individuals from gaining physical access to networking equipment and compromising the confidentiality of the network. This can be achieved through measures such as biometric systems, access cards, and other physical security devices.



How Does Network Security Work?

2. Technical Network Security: It primarily focuses on protecting the data stored in the network or data involved in transitions through the network. This type serves two purposes. One is protected from unauthorized users, and the other is protected from malicious activities.

How Does Network Security Work?

3. Administrative Network Security: This level of network security protects user behavior like how the permission has been granted and how the authorization process takes place. This also ensures the level of sophistication the network might need for protecting it through all the attacks. This level also suggests necessary amendments that have to be done to the infrastructure.

Common Threats: Virus, Malware, Phishing, Hacking

- Discussed in Cyber Security Lecture

Quiz

What is the main goal of network security?

- A. To enhance network speed
- B. To protect data integrity, confidentiality, and availability
- C. To improve user experience
- D. To reduce operational costs

Quiz

Which of the following is a common method to prevent unauthorized physical access to network equipment?

- A. Implementing access control lists
- B. Using biometric security systems
- C. Increasing bandwidth
- D. Updating software regularly

Quiz

What does network segmentation aim to achieve?

- A. Faster data transmission
- B. Enhanced security through traffic categorization
- C. Simplified network management
- D. Increased user permissions

Quiz

In the context of network security, what is the function of an Intrusion Prevention System (IPS)?

- A. To speed up network traffic
- B. To monitor and block malicious activities
- C. To provide user access control
- D. To encrypt data transmissions

Quiz

Which of the following best describes the purpose of email security?

- A. To enhance email delivery speed
- B. To protect email accounts from unauthorized access and threats
- C. To facilitate easier communication
- D. To reduce spam messages

Research

- Network Security vs Cybersecurity vs Cloud Security

Summary

- Network security protects data and users.
- Common threats include viruses, phishing, and ransomware.
- Tools like firewalls, encryption, and VPNs improve safety.
- Follow safe practices and ethical communication online.